

Project 1

Designing a Jig for a Cylindrical Part

MEC 816: Group Project

April 2025

1.0 Drawings

1.1 Deliverable 1:

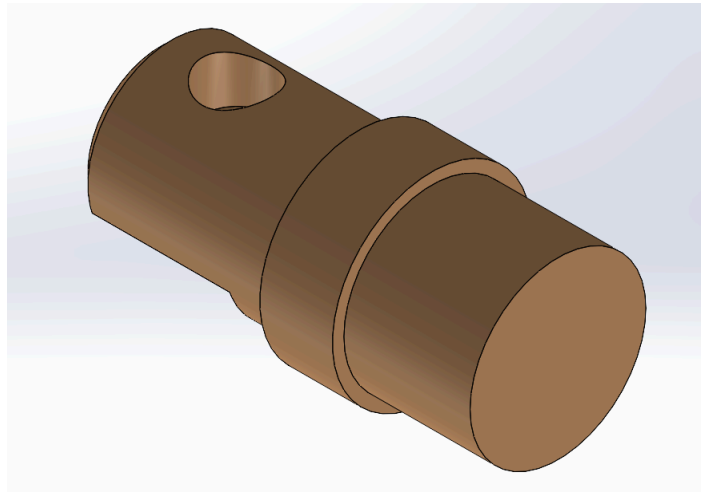


Figure 1: Final Part CAD

1.2 Deliverable 2:

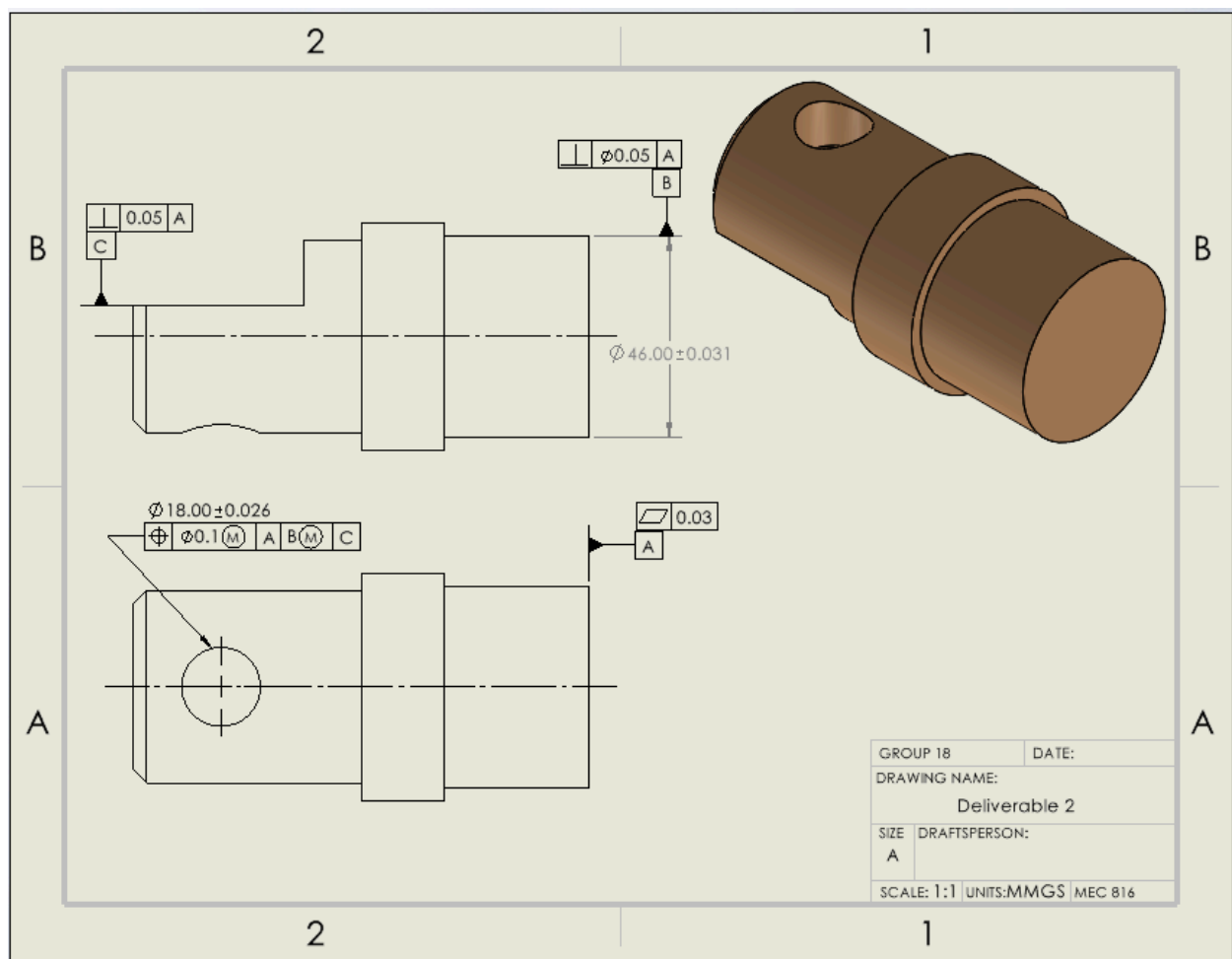


Figure 2: GD&T Drawing of Part

2.0 Locating System

Datum A

The part of the jig design that incorporates datum A was a flat-machined wall. This wall was utilized to constrain the part's translational motion in the x-axis. This machined wall was constructed using a built up body, including 2 dowels and two cap screws. The part becomes flush with the machined wall, representing datum A when the set screw locks the surface onto the datum A wall.

Datum B

Since datum B is perpendicular to datum A, a V-block was used to ensure consistent alignment of the shaft's central axis, making it ideal as a secondary datum. The V-block minimizes part movement and ensures proper perpendicularity and concentricity for features referenced. The V-block provides a stable and repeatable way to locate and support the cylindrical surface, and the two angled surfaces of the block create line contact with the cylindrical feature, effectively constraining the part in the horizontal plane.

Datum C

Datum C uses 2 support pins to act like a locating feature. The pins remove rotation in the X axis, finally locking the part in space; removing all 6 DoF. Using two pins instead of one helps in evenly distributing the load and provides better stability, minimizing the chances of misalignment due to wobbling or tipping.

3.0 Clamping System

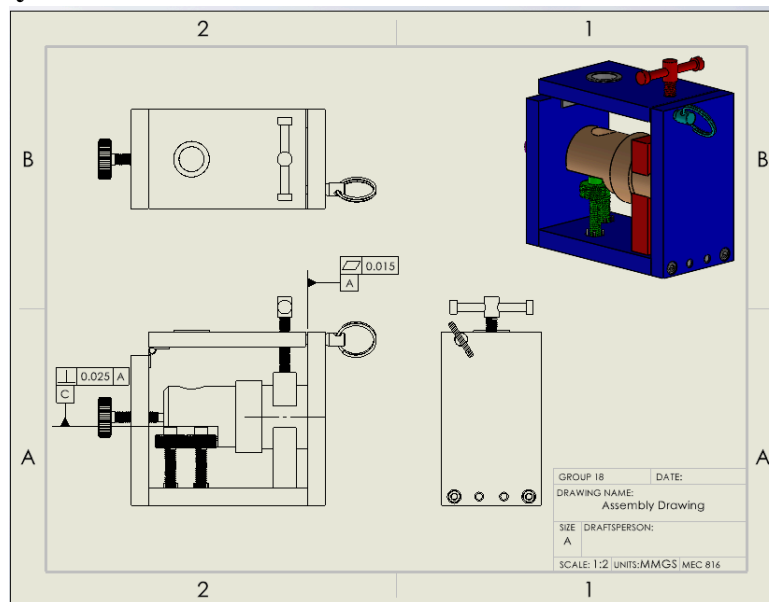


Figure 3: Jig Assembly Drawing

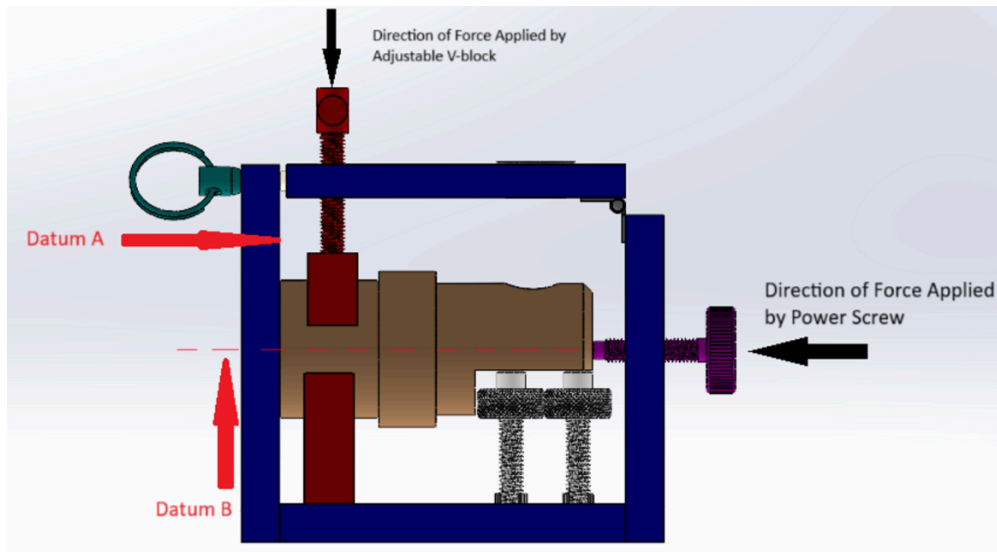


Figure 4: Direction of Forces Applied

An M8x1.25 set screw [8] is used to push the part onto the Datum A simulator, and the size was chosen to have as much surface area contact on the part while loading the force in the center of its axis. Additionally the V-block on the top uses a power screw to clamp the part between the top and bottom V-blocks, analogous to a bench vice. The power screw serves as a method to provide a preload to the part as required, hence removing 4 DOF.

4.0 Tool Guidance

4.1 Bushing Type

For the circled red feature in the project manual, a bushing needed to be selected for drilling purposes. The group selected an 18mm H-type press fit bushing from McMaster-Carr. This type of bushing was selected due to its flanged head, which allows the bushing to withstand drilling pressure without pressing through jig plates[1]. Ultimately, this can help protect the jig from drilling damage/wear for long periods. This part may easily be replaced from McMaster Carr if needed, as it is an off-the-shelf component.

4.2 Bushing Size and Location

When comparing to design specifications on McMaster-Carr, the following calculations were compared to the values listed on the product's website. Therefore, this bushing was further justified for its extended drill guide application.

$$\text{Internal Diameter} = d + \text{clearance} = 18\text{mm} + 0.025\text{mm} = 18.025\text{mm}$$

$$\text{External Diameter} = 1.5 \times d = 1.5 \times 18\text{mm} = 27\text{mm}$$

$$\text{Head Diameter} = 2 \times d = 2 \times 18\text{mm} = 36\text{mm}$$

$$\text{Head Thickness} = 0.625 \times d = 0.625 \times 18\text{mm} = 11.25\text{mm}$$

$$\text{Length of the Bush} = 2.5 \times d = 2.5 \times 18\text{mm} = 45\text{mm}$$

5.0 Tolerance Calculations

5.1 Geometric Tolerancing

Datum A on the part has a flat tolerance applied to it; hence, applying the 50% rule the flatness tolerance applied to datum A on the jig was as follows:

$$T_A = (0.03)(0.5) = 0.015\text{mm}$$

Datum C on the part has a perpendicularity tolerance applied to it, Applying the 50% rule the perpendicularity tolerance applied to datum C on the jig was as follows:

$$T_c = (0.05)(0.5) = 0.025mm$$

6.0 Material selection

MIC-6 cast aluminum was chosen for datum A's fixture plate because it offers great flatness, stability, and is easy to machine. Since our fixture relies on a flat surface for Datum A, we needed a material that wouldn't warp or shift over time. MIC-6 is stress-relieved during casting through a built-in heat treatment process, which helps prevent internal stresses from causing distortion [5]. That means we don't need any additional heat treatment after machining, and we can count on the surface staying flat and accurate. It's also lightweight, which makes handling the fixture easier without sacrificing precision[6].

The V-blocks were also custom-made and were manufactured out of 1018 mild steel because it was cost-effective, widely available, and was known for its excellent machinability and surface finish. These properties make it ideal for accurately machining the V-grooves needed to support the cylindrical part. However, since 1018 is a low-carbon steel, it has relatively low hardness and wear resistance in its untreated state. To enhance durability, hardening was recommended. This process forms a thin, hardened surface layer while preserving a tough and ductile core, improving the material's resistance to wear and deformation under repeated use. 1018 steel is often used for parts that require strength, ductility, and case hardening [7]. Heat treating expands its range of applications, making it more suitable for fixture components like V-blocks in moderate-duty setups.

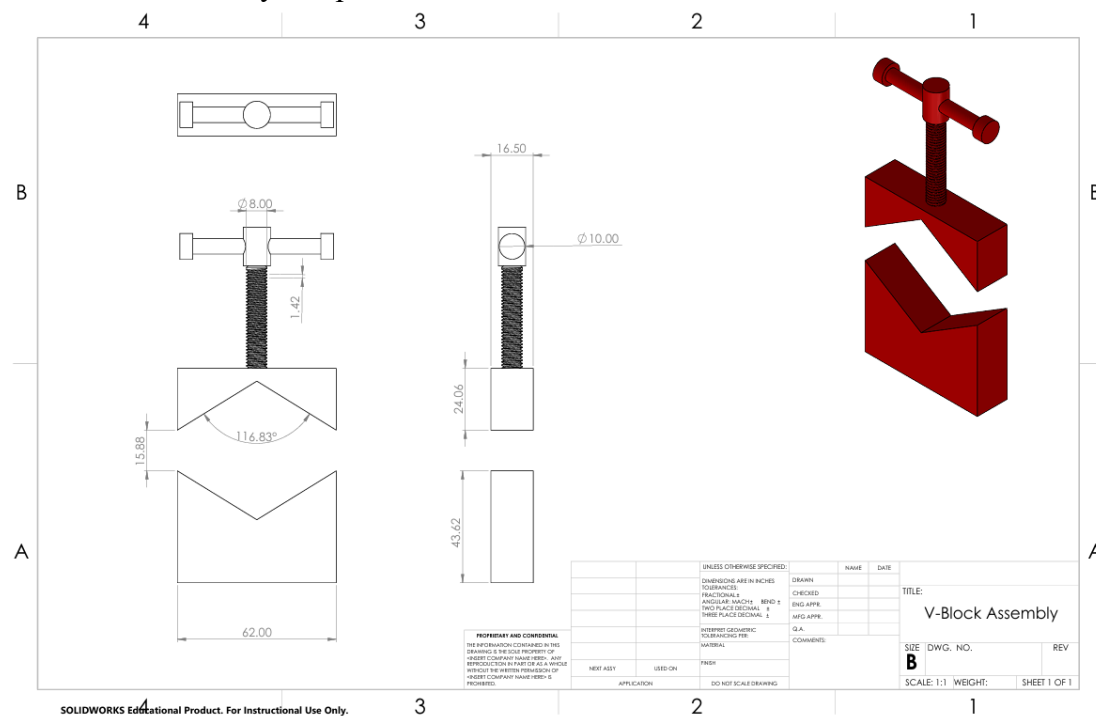


Figure 5: V-block part drawing

8.0 References

- [1] McMaster-Carr, “Metric Press-Fit Drill Bushings with Head,” *McMaster-Carr*, Accessed: Apr. 8, 2025. [Online]. Available: <https://www.mcmaster.com/products/bushings/metric-press-fit-drill-bushings-with-head/system-of-measurement~metric/id~18-mm/>
- [2] Carr Lane, “Knurled Screw Jacks,” *Carr Lane Manufacturing Co.*, Accessed: Apr. 8, 2025. [Online]. Available: <https://www.carrlane.com/product/work-supports-rest-buttons/screw-jacks/knurled-screw-jacks>
- [3] Carr Lane, “Key Inserts, Heavy Duty, Steel, Inch,” *Carr Lane Manufacturing Co.*, Accessed: Apr. 8, 2025. [Online]. Available: <https://www.carrlane.com/product/threaded-inserts/key-inserts-heavy-duty/key-inserts-heavy-duty-steel-inch>
- [4] McMaster-Carr, “18-8 Stainless Steel Knurled-Head Thumb Screw, 1/4"-20 Thread Size, 3/4" Long,” *McMaster-Carr*, Accessed: Apr. 8, 2025. [Online]. Available: <https://www.mcmaster.com/98381A537/>
- [5] Alcoa (now Arconic). MIC-6 Cast Plate Technical Data Sheet.
- [6] Clinton Aluminum. “MIC-6 Aluminum Cast Plate.” <https://www.clintonaluminum.com/products/aluminum/mic-6/>
- [7] MatWeb. *AISI 1018 Steel, cold drawn*. <https://www.matweb.com/search/datasheet.aspx?bassnum=M1018A>
- [8] McMaster-Carr, "Alloy Steel Cup-Point Set Screw," *McMaster-Carr*, [Online]. Available: <https://www.mcmaster.com/98014A644/>. [Accessed: Apr. 9, 2025].
- [9] McMaster-Carr, "Steel Piano Hinge without Holes," *McMaster-Carr*, [Online]. Available: <https://www.mcmaster.com/16175A12/>. [Accessed: Apr. 9, 2025].
- [10] Carr Lane Manufacturing Co., "Detent Pins," *Carr Lane*, [Online]. Available: <https://www.carrlane.com/product/ball-lock-pins-quick-release-pins/detent-pins/detent-pins>. [Accessed: Apr. 9, 2025].

9.0 Appendix
9.1 Drawings

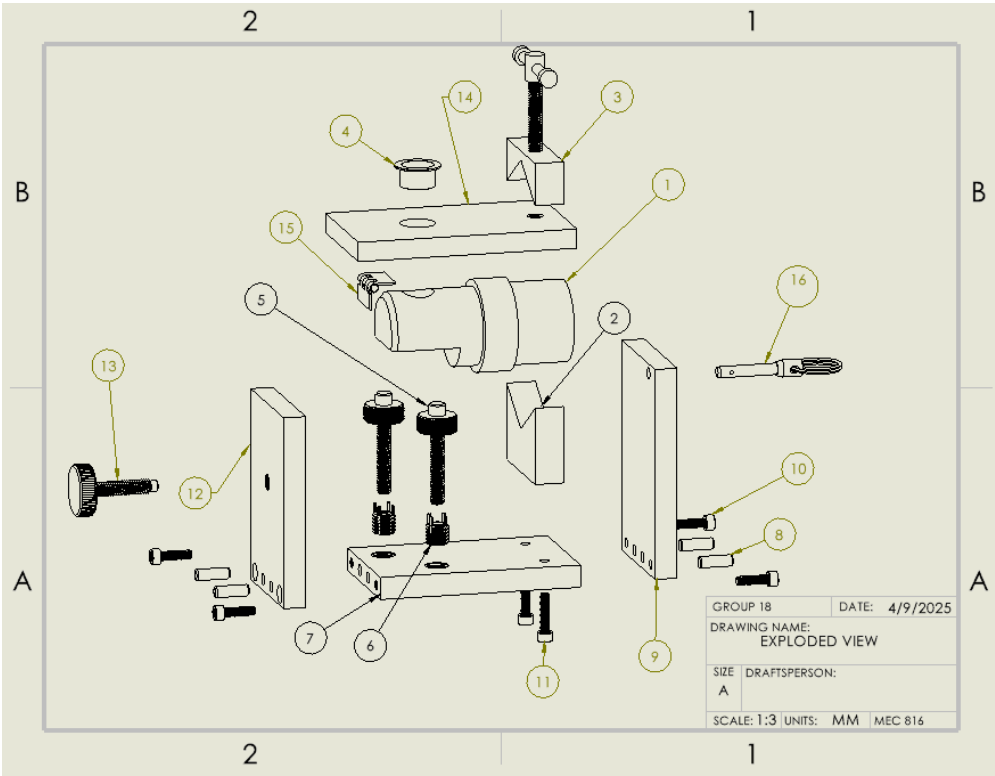


Figure 5: Jig Exploded View

2		1	
ITEM NO.	PART NUMBER	DESCRIPTION	QTY.
1	MEC 816 - Group 18 - Jig Design Project	Part 18	1
2	Custom V Block Bottom		1
3	Custom V Block Top _NEW		1
4	6659K264	Oil-Embedded 841 Bronze Flanged Sleeve Bearing	1
5	cl-0-ksj	Steel Datum C Pin	2
6	cl-51618-ks_01	Threaded Insert	2
7	Jig base		1
8	98381A540	Dowel Pin	4
9	Datum A	Datum A vertical wall	1
10	91290A238	Alloy Steel Socket Head Screw	4
11	91290A248	Alloy Steel Socket Head Screw	2
12	Vertical wall		1
13	98014A644	Stainless Steel Knurled-Head Extended-Tip Thumb Screw	1
14	Top Plate		1
15	16175A12	Steel Hinge without Holes	2
16	cl-4-dep-1_00-s_01	Detent Pin	1

GROUP 18 DATE: 4/9/2025
DRAWING NAME: BOM
SIZE: A DRAFTSPERSON:
SCALE: 1:3 UNITS: MM MEC 816

Figure 6: Bill of Materials

Project 2

Designing a Functional Gage and GO/NO GO Gage for Part

MEC 816: Group Project

April 2025

1.0 Updated Multiview GD&T Drawing of Part

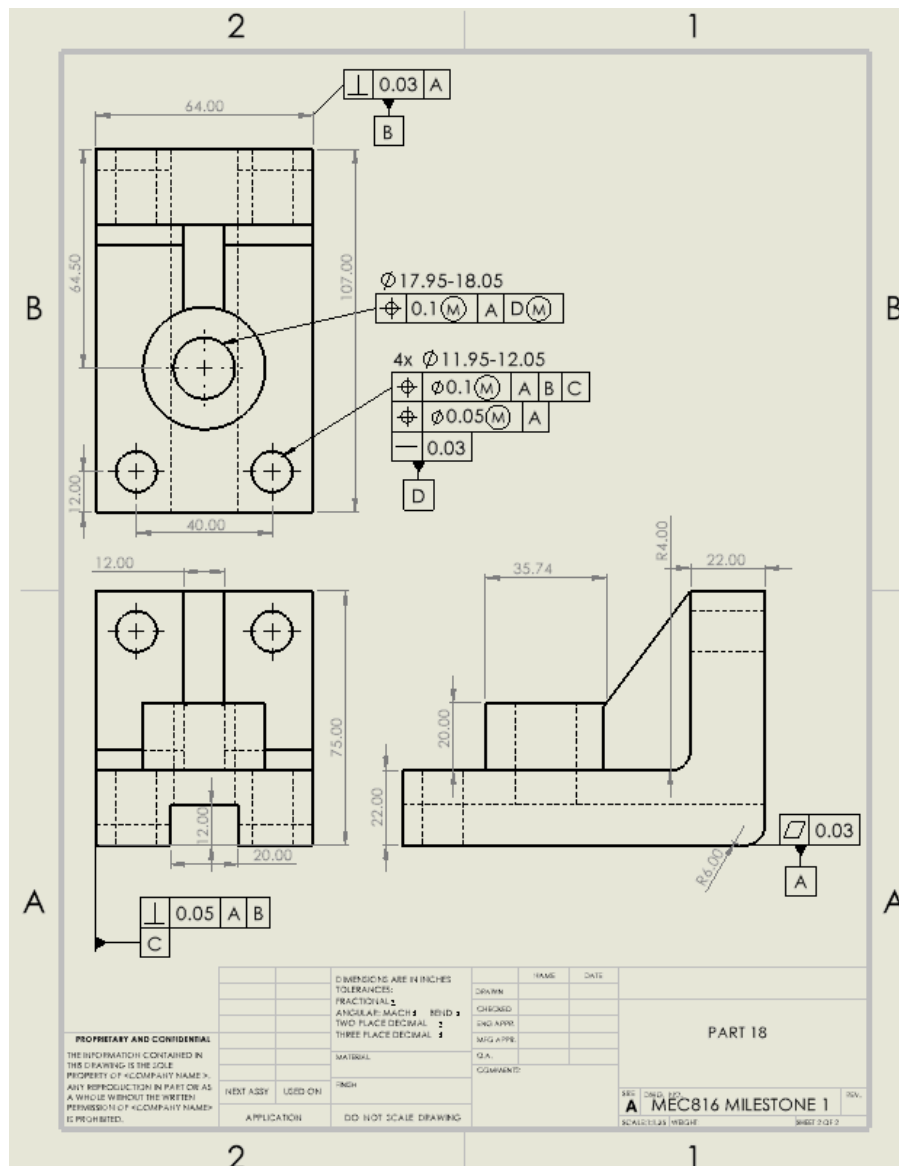


Figure 1: Updated GD&T drawing of the assigned part.

1.2 Functional Gage Selection for Milestone 3

Gage 1 from Milestone 2 was chosen for Milestone 3. This gage verified the pattern locating tolerance (PLTZF) of the composite tolerance for the holes.

2.0 Go/No-Go Gage Detailed Design

2.1 Tolerance Calculations

2.1.1 18mm Hole Go/ No-Go Gage for Size Tolerance

Tolerance for the 18mm hole= $18.05 - (17.85 - 0.1) = 0.3$

Therefore the tolerance is ± 0.15

Using the 10% rule: $= \pm 0.15 \times 0.1 = \pm 0.015 \text{ mm}$

Absolute (pessimistic approach):

The tolerance for the go gage side goes to the plus side, whereas for the no-go gage the tolerance does to the minus side. The shaft length for the go gage side is the entire length of the part that is being gages, therefore it is 42 mm. The no-go gage does not require full engagements, therefore the length chosen for the shaft is 20 mm.

Go Gage Diameter: $17.75 + 0.03 \text{ mm}$

No go Gage Diameter: $18.05 - 0.03 \text{ mm}$

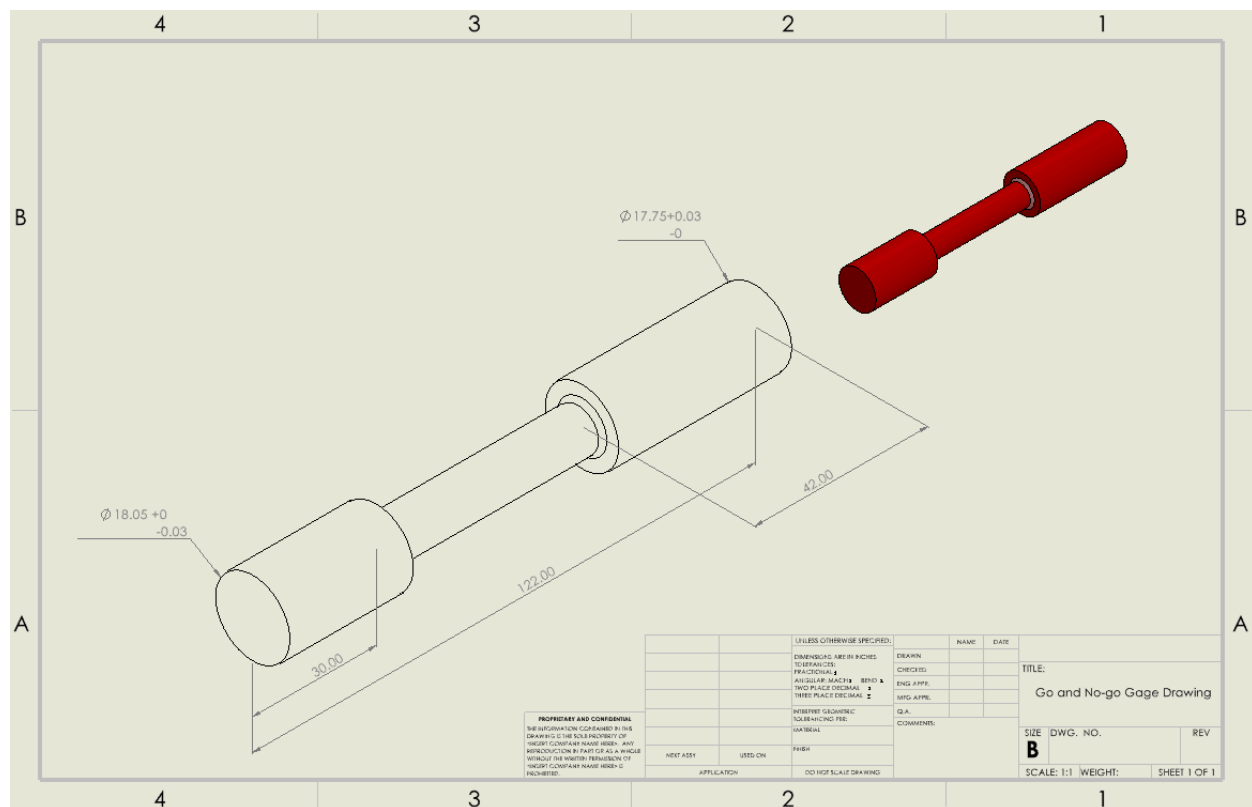


Figure 2: Go Gage/No-go Gage Drawing

3.0 Functional Gage Detailed Design

3.1 Material Selection

When developing a push pin design and functional gauge block for the part, selecting appropriate materials was essential to ensure durability, precision, and ease of use. The chosen materials must withstand repeated insertion and removal while maintaining dimensional stability for accurate gauging. For the functional gauges, 1018 mild steel was selected due to its excellent machinability, good weldability, and sufficient strength for moderate load applications [2]. This low carbon steel offers a balance of toughness and cost-effectiveness, making it ideal for gauge applications that require precision but do not experience extreme wear.

On the other hand, Grade D2 tool steel was chosen for the push pins due to its high hardness, exceptional wear resistance, and ability to retain a sharp edge over prolonged use [3]. D2 tool steel, a high-carbon, high-chromium air-hardening steel, provides superior durability, making it suitable for repeated engagement with the gauge wall.

Additionally, alternative materials such as Delrin and nylon were considered for their lightweight properties and wear resistance, but metals were ultimately preferred for their strength and longevity in this high-precision application [4], [5]. Balancing these properties strength, wear resistance, manufacturability, and cost-effectiveness ensures optimal functionality and reliability of the push pins and gauge wall in their intended application.

3.2 Tolerance calculations

3.2.1 Flatness Tolerance for Datum A

To find the flatness tolerance for the datum A simulator, the 10% rule is applied, in which every geometric tolerance is multiplied by 10% to obtain the gauge tolerance [cite].

$$Flatness_{Datum A} = 0.03 * 0.1 = 0.003mm$$

This indicates that the flatness tolerance on the top surface of the gauge is 0.003mm.

3.2.2 Perpendicularity Tolerance for Datum B

To determine the perpendicularity tolerance for the simulator of datum B on the functional gauge, the 10% rule was applied.

$$Perpendicularity_{Datum B} = 0.03 * 0.1 = 0.003mm$$

3.2.3 Push pin Design Calculation

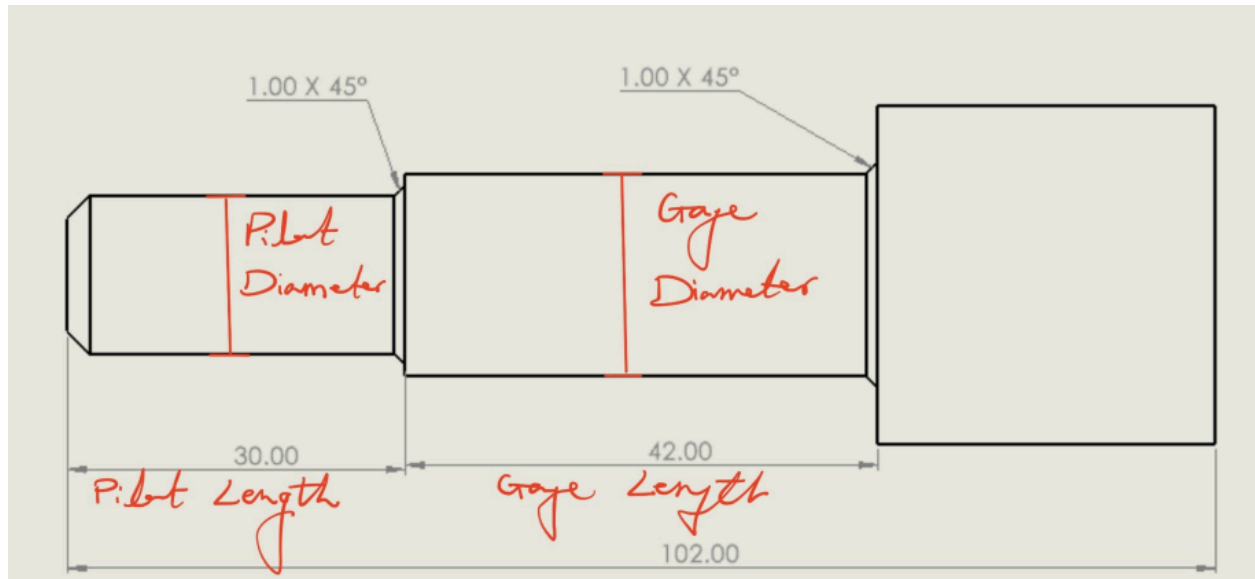


Figure 3: Push-Pin Diagram

The gauge diameter was found in milestone 2 as the virtual condition using the IT9 tolerance standard. This standard states that for holes of diameter 10mm to 18mm, the geometric tolerance will be $\frac{0.043}{2} = 0.0215 \text{ mm}$. Using this, the virtual condition of the can be found below:

$$\begin{aligned} VC &= MMC - \text{geometric tolerance} \\ VC &= 17.95mm + 0.0215mm \\ VC &= 17.9715mm \approx 17.98mm \end{aligned}$$

From figure 1, it can be seen that the size tolerance for the large hole is $8.05 - 17.95 = 0.1\text{mm}$. Applying the 10% rule to this, a total tolerance of 0.01mm is obtained for the go/no-gauge.

The gauge length is determined by setting a length which allows the pin to fully go through the part. The following calculation was done to determine the gauge length based on Figure 1:

$$\text{Gauge Length} = 20\text{mm} + 22\text{mm} = 42\text{mm}$$

The 20mm represents the circular extrusion of the large hole, and the 22mm represents the base thickness of the part.

The pilot length was chosen based on the bushing that was selected, which had a length of 30mm, hence requiring the pilot to also be 30mm. Also, the pilot diameter was chosen to be 13.84mm due to the ID of the selected off-the-shelf bushing.

4.0 Final Design

4.1.1 Part drawings

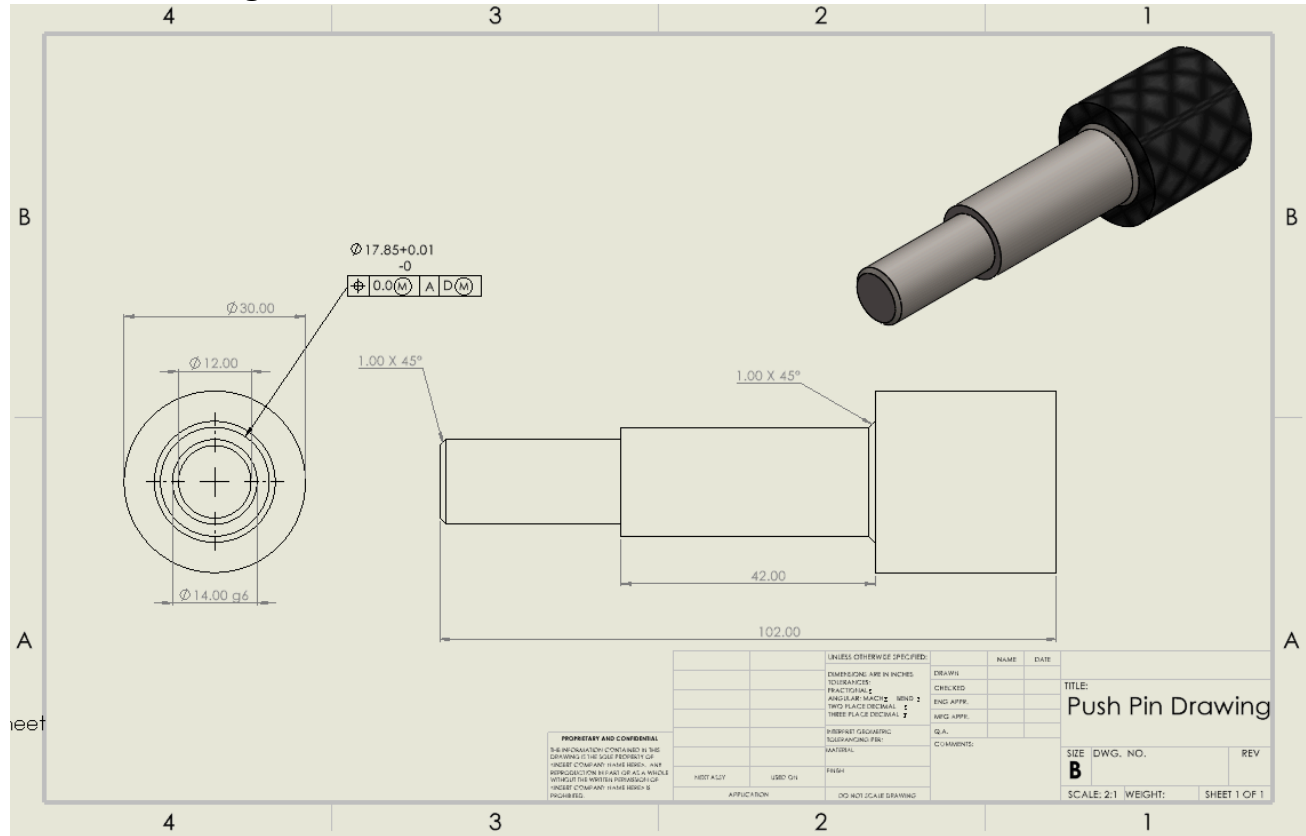


Figure 4: Drawing of Slip Pin for 18 mm Hole

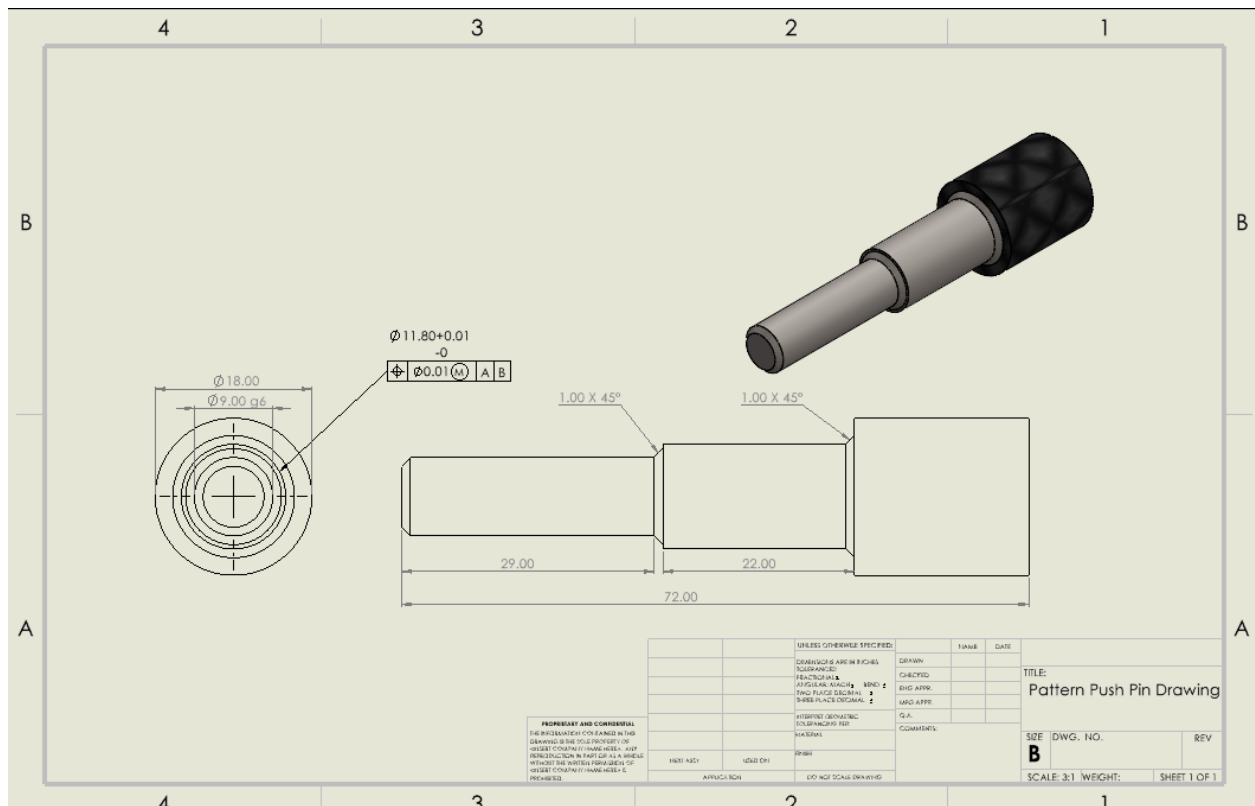


Figure 5: Drawing of Slip Pin for Pattern of Holes

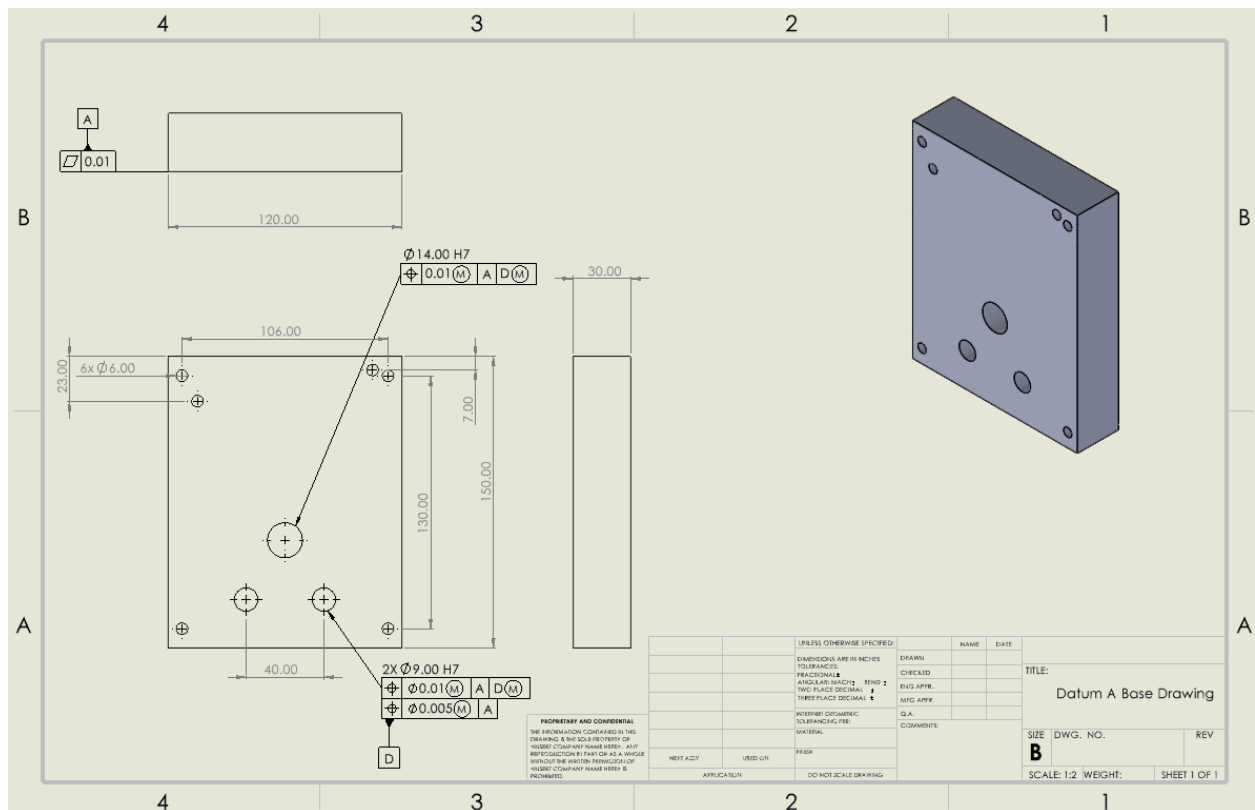


Figure 6: Drawing of Datum A Simulator

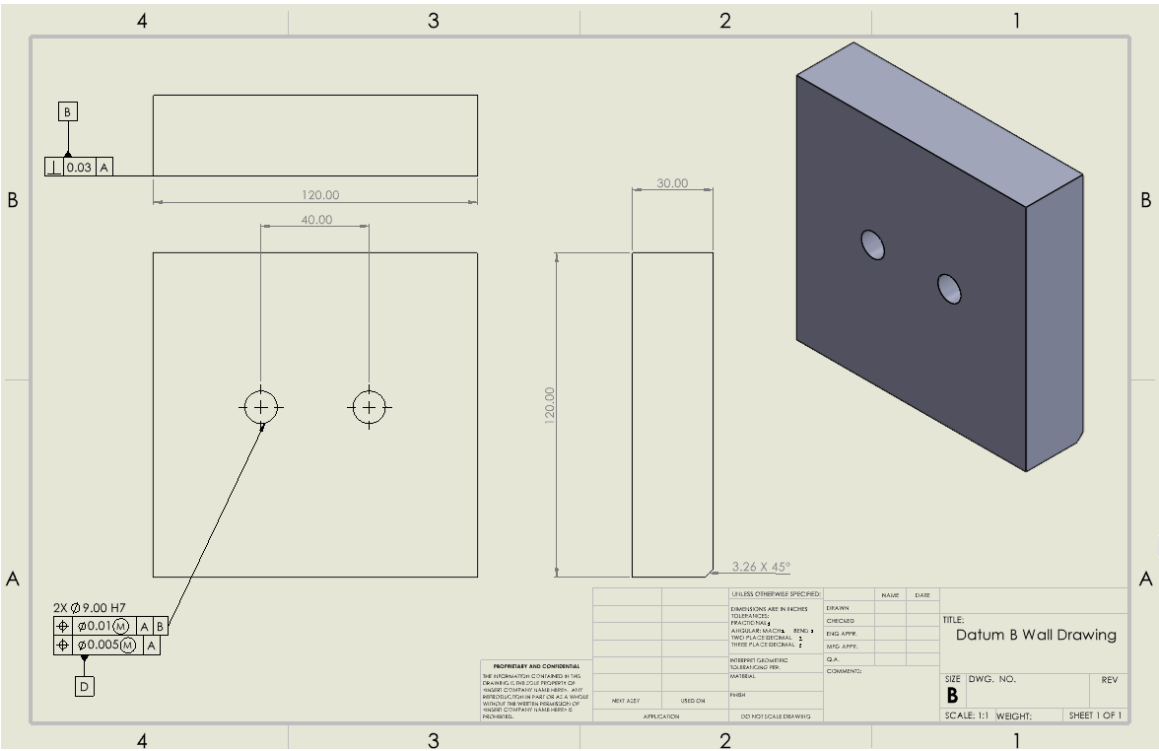


Figure 7: Drawing of Datum B simulator

4.1.2 Assembly drawings

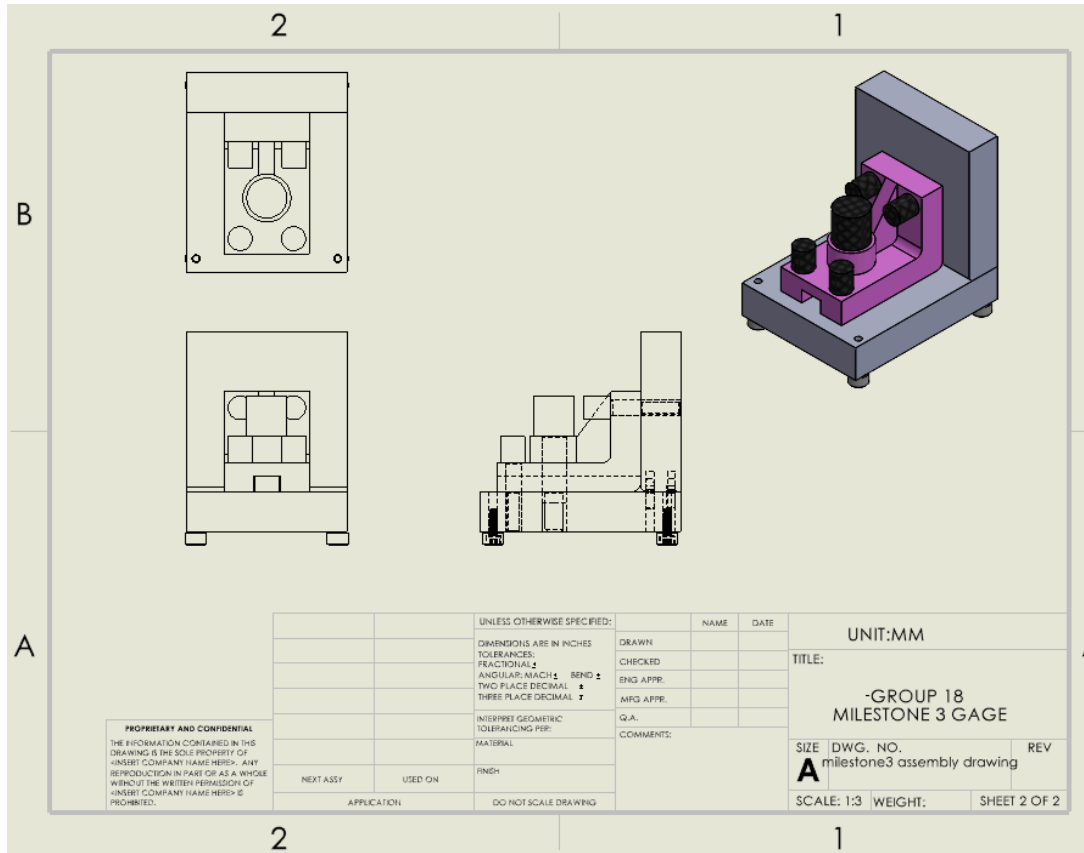


Figure 8: Overall Assembly Drawing

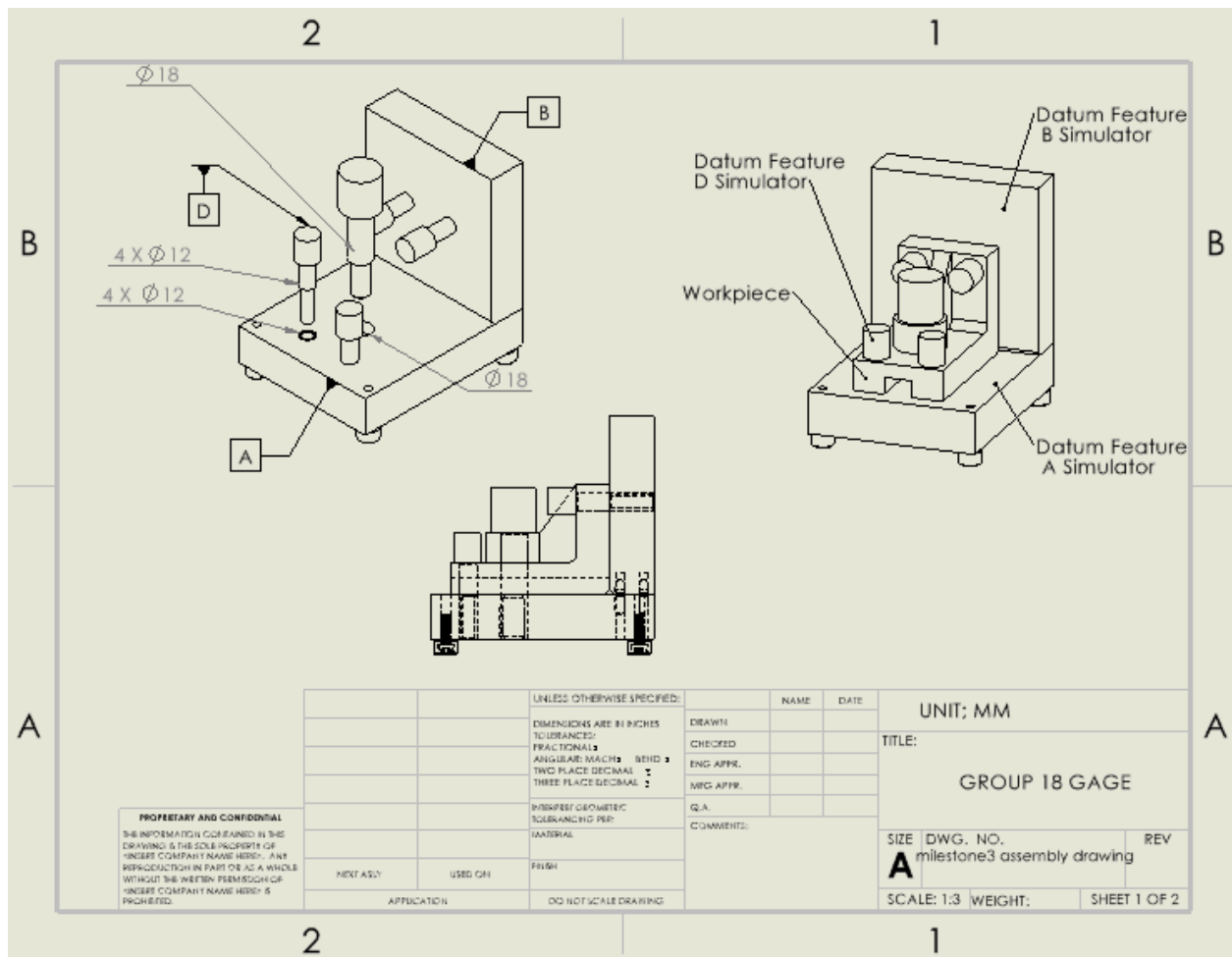


Figure 9: Overall Assembly Drawing with Datums

4.1.3 Exploded view drawings

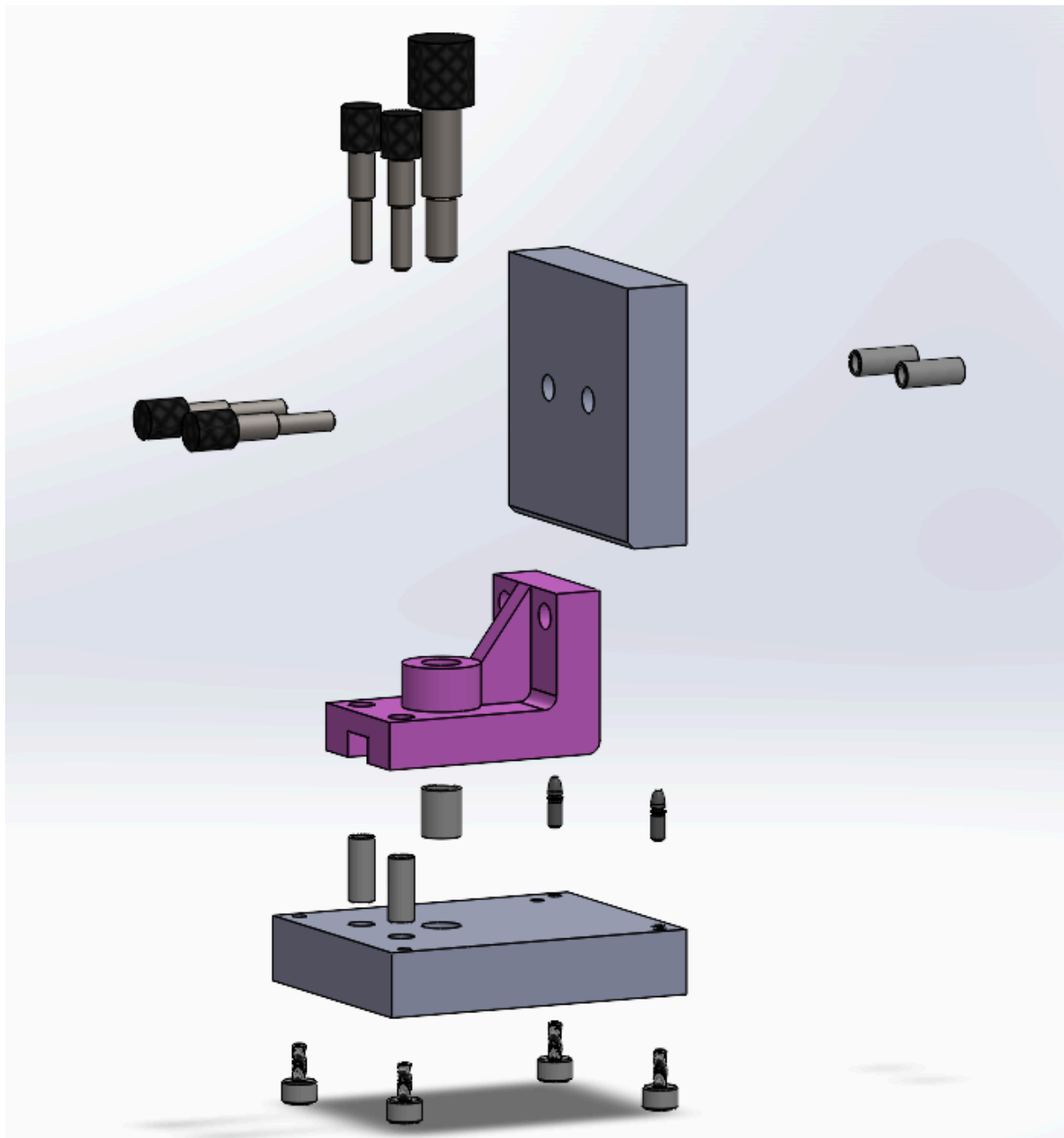


Figure 10: Gage Assembly

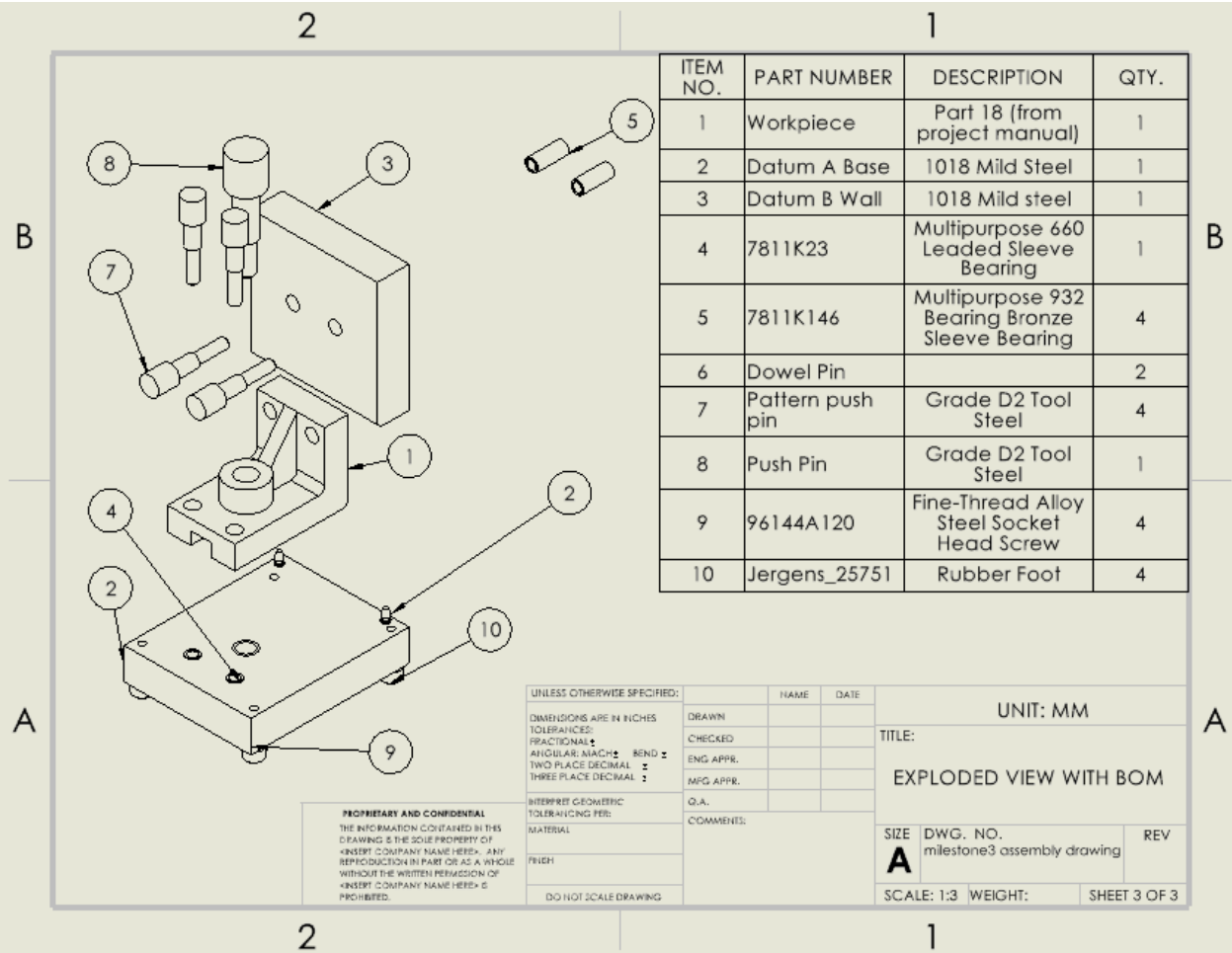


Figure 11: Exploded View with BOM

4.3 Functional Gage Operator Instructions

The operator instructions for assembly of functional gage are as listed below:

1. Press fit dowel Pins into Datum A Base.
2. Line up Datum B Wall with Dowel Pins and slot into place.
3. Slot 12mm OD bushings into 4 pattern of holes features (Datum D).
4. Slot 18mm OD bushing into 18mm hole.
5. Position workpiece on functional gage.
6. Align workpiece pattern of hole features with 4 12mm pattern of holes on functional gage.
7. Use handheld 12mm diameter push pin GO gage and slot into 4 coaxial holes.
8. Insert handheld 18mm diameter push pin GO to gage to 18mm hole in part to check coaxial size.
9. Remove all push pins from bushings and gage walls.
10. Unclamp part and remove from functional gage.

Part must pass GO-NO-GO size checks.

Part must fit on gage and push pins must fit through holes.

5.0 References

- [1] "Ghamsempoor, A. "Project 1, Milestone 2", Topics in Manufacturing Engineering course materials, 2025
- [2] "1018 Mild Steel: Properties and Uses," *AZO Materials*, Available: <https://www.azom.com>
- [3] "D2 Tool Steel: Properties, Applications, and Heat Treatment," *ASM International*, Available: <https://www.asminternational.org>
- [4] Dupont, *Delrin Product and Properties Guide*, Distrupol. Available: https://www.distrupol.com/Delrin_Product_and_Properties_Guide_Dupont.pdf
- [5] Rapid Direct, "Delrin vs Nylon: Differences, Advantages & Applications," *Rapid Direct Blog*. Available: <https://www.rapiddirect.com/blog/delrin-vs-nylon/>
- [6] Carr Lane Manufacturing Co., "Round locating pins, press-fit type," *Carr Lane*, Accessed: Mar. 24, 2025. [Online]. Available: <https://www.carrlane.com/product/locating-pins/locating-pins/round-diamond-locating-pins-press-fit-type/round-locating-pins-press-fit-type>
- [7] McMaster-Carr, "Pattern hole bushing," *McMaster-Carr*, Accessed: Mar. 24, 2025. [Online]. Available: <https://www.mcmaster.com/7811K132/>
- [8] McMaster-Carr, "Big center hole bushing," *McMaster-Carr*, Accessed: Mar. 24, 2025. [Online]. Available: <https://www.mcmaster.com/products/bushings/sleeve-bearings-3~/multipurpose-sleeve-bearings-8/shaft-diameter8-mm/shaft-diameter14-mm/od18-0508-mm/length22-mm/>
- [9] Jergens Inc., "Flat foot, M6 x 20 metric (jig feet)," *Jergens Inc.*, Accessed: Mar. 24, 2025. [Online]. Available: <https://www.jergensinc.com/en/product/stock-check/flat-foot-m6-x-20-metric>
- [10] McMaster-Carr, "Socket head screws, M6 x 0.75 mm thread (96144A120)," *McMaster-Carr*, Accessed: Mar. 24, 2025. [Online]. Available: <https://www.mcmaster.com/catalog/131/3557/96144A120>